

HEWEN SHEN

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EDUCATION

Northeastern University | Seattle, WA

Sep 2024 – May 2027 (Expected)

Master of Science in Computer Science - Align (GPA:4.0)

WORK EXPERIENCE

Prospect Equities | Full Service Real Estate Brokerage

Chicago, IL

Frontend Developer Intern

Jan 2026 – April 2026

- Expanded an internal transaction management system by designing and implementing **pre-payment commission workflows from the ground up** in **React**, supporting deal-specific agent split calculations, commission confirmation, and tracking to increase deal lifecycle coverage by ~30%.
- Built reusable **funds intake/release components** with **React.js** and **Ant Design**, merging and normalizing intake/release data from two **RESTful APIs** into a unified, editable transaction timeline, with **multipart receipt uploads** and **consolidated PDF download of all funds records per deal**.
- Integrated frontend **funds-release operations** with RESTful APIs supporting **PostgreSQL persistence**, **AWS S3 receipt storage**, **MongoDB Outbox**, and **Kafka-based asynchronous notifications**, covering the flow from funds submission through agent notification.
- Enhanced the **deal-progress steps** in **React.js** by introducing a new **commission confirmation milestone** and implementing state-based logic to automatically trigger the corresponding modal, reducing redundant user interactions by ~20%.

PROJECTS

Online Learning Microservices Platform

May 2026 – Aug 2026

- Built a **reusable distributed locking framework** with **Spring AOP**, **Redisson**, and **custom annotations** to decouple concurrency control from business logic, applying Factory and Strategy patterns to support 5 configurable lock types and 3 lock-acquisition strategies.
- Redesigned the concurrent **coupon-claiming workflow** with **Redisson distributed locks** and **atomic conditional stock updates**, improving successful allocation under load from **21/100 to 100/100** without overselling or per-user limit violations; used **Redis BitMaps** for atomic redemption tracking.
- Optimized **high-concurrency learning-progress persistence** using **Redis** and delayed tasks to buffer and coalesce frequent progress updates, reducing **MySQL write frequency and volume by ~95%** while preserving completion-state consistency.
- Implemented a seasonal **learning-points leaderboard** with **Redis Sorted Sets** for real-time rankings and season-partitioned **MySQL** tables for historical data, using a **distributed job scheduler with instance-level sharding** to parallelize end-of-season persistence across service instances.

E-Commerce Microservices Platform

Jul 2025 – Nov 2025

- Architected an **order-to-payment workflow** across Spring Cloud microservices, routing and authenticating requests through an **API Gateway** while using **Redis and Lua** for atomic inventory reservation and **RocketMQ** for asynchronous event propagation.
- Implemented **RocketMQ transactional messaging** to propagate payment-success events across trade, inventory, and cart services, **avoiding dual-write inconsistencies** and enforcing idempotent consumption with **deduplication tables keyed by order and event type**.
- Built a **timeout-handling and compensation workflow** that reconciles inconsistent payment/order states through a compensation table and scheduled batch verification, canceling unpaid orders and restoring reserved stock only after confirming true payment status.
- Developed an **Elasticsearch product search service** supporting multi-field filtering and aggregation to improve product discovery across the catalog.

SKILLS

- Languages:** Java, Python, Go, JavaScript/TypeScript, SQL, HTML/CSS
- Frameworks & Libraries:** Spring Boot, Spring Cloud, MyBatis-Plus, React, Vue, Express.js, JUnit
- Databases & Storage:** MySQL, Redis, MongoDB, Elasticsearch
- Message Queues:** RocketMQ, Kafka, RabbitMQ
- Cloud & DevOps:** AWS (EC2, S3, Lambda), Docker, Nginx, Git, GitHub Actions, Jenkins
- AI Development Tools:** Codex, Cursor